

Adaptive Forgetting Factor for Drifting Inverter-Based Resource Penetration in Model-Based Protection Systems

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Abstract—The Self-Adaptive Model-Based Protection System (SAMBPS) uses a Levenberg–Marquardt (LM) zone-model estimator to adapt relay settings to the instantaneous inverter-based resource (IBR) current ratio k_{ibr} . Existing SAMBPS implementations assume k_{ibr} is approximately stationary over the estimation window. In practice, solar photovoltaic output follows a diurnal ramp, wind generation executes rapid fluctuations, and topology changes create step discontinuities — all of which cause k_{ibr} to drift on timescales of seconds to hours. A fixed exponential forgetting factor $\lambda = 0.995$ (effective window $N_{eff} = 200$ samples) introduces a 63 ms tracking lag that translates to $P_{FA} = 0.032$ during cloud transients and P_D degradation of up to 16% under sustained ramps. We propose an adaptive forgetting factor $\lambda(t)$ driven by a CUSUM-based drift detector applied to the sequence of LM estimates $\hat{k}_{ibr}$. A logistic transition law interpolates between $\lambda_{min} = 0.970$ ($N_{eff} = 33$, fast) during detected drift and $\lambda_{max} = 0.995$ ($N_{eff} = 200$, low noise) during stationary conditions. A stability proposition proves that $N_{eff} \geq 2n_{free}$ at all times, bounding the normalised Jacobian condition number $\kappa_n < 30$ throughout. Monte Carlo validation over 20 000 trials across four drift scenarios (solar diurnal, cloud transient, wind random walk, topology step) shows: tracking RMSE reduced $3.4\times$ (0.062 to 0.018 pu), P_{FA} reduced to zero for all scenarios, and $P_D \geq 0.981$ versus 0.789–0.862 for fixed λ .

Index Terms—adaptive protection, forgetting factor, inverter-based resources, Levenberg–Marquardt, recursive estimation, model-based protection, CUSUM drift detection

I. INTRODUCTION

The proliferation of inverter-based resources (IBRs) in distribution and sub-transmission networks has fundamentally altered the fault current environment seen by protection relays. Unlike synchronous generators, whose fault current is determined by machine reactances and decays predictably on a 30–200 ms sub-transient timescale, IBRs contribute a current controlled by firmware: magnitude-limited to $[1.1, 1.5] I_{rated}$ (GFL) or $[1.5, 2.5] I_{rated}$ (GFM), and phase-angle commandable over a $\pm 90^\circ$ range under IEEE 1547-2018 and IEEE 2800-2022 requirements [1], [2].

The IBR current ratio k_{ibr} — the per-unit fraction of the zone rated current supplied by IBRs at fault inception — is the single most important parameter linking IBR physics to relay algorithm behaviour in the SAMBPS framework [3]. Protection schemes calibrated for $k_{ibr} = 0$ (pure synchronous generation)

suffer well-documented failures at $k_{ibr} \geq 0.3$: differential bias misoperation, distance-relay under-reach, and overcurrent time-multiplier mismatch [4], [5]. SAMBPS addresses this by estimating k_{ibr} in real time from the measured waveforms and updating relay settings on a sub-cycle (20 ms) basis using a Levenberg–Marquardt zone-model estimator [3].

The stationarity assumption and its breakdown. All SAMBPS implementations reported to date assume that k_{ibr} is approximately constant within the LM estimation window, typically spanning 1–2 fundamental cycles (20–40 ms) [6]. This assumption holds during a fault event, where the IBR control loop responds within 2–5 ms and then maintains a constant current [7]. Between fault events, however, k_{ibr} drifts continuously:

- *Solar diurnal ramp*: typical irradiance rises from 0.05 to 0.52 pu over 4 hours ($\dot{r}_k \approx 2 \times 10^{-3}$ pu/cycle);
- *Cloud transient*: partial shading drops k_{ibr} by 0.30–0.40 pu in under 30 s ($\dot{r}_k \approx 10^{-2}$ pu/cycle);
- *Wind random walk*: k_{ibr} executes a correlated random walk with $\sigma = 0.05$ pu/min;
- *Topology step*: a capacitor bank switching or network reconfiguration causes a step change in k_{ibr} of up to 0.20 pu in one cycle ($\dot{r}_k \approx 0.20$ pu/cycle).

Between fault events, the LM estimator continuously refines its initial parameter estimate from the load-flow operating point, and a stale $\hat{k}_{ibr}$ delays the protection response when the next fault occurs. With a fixed forgetting factor $\lambda = 0.995$ ($N_{eff} = 200$ cycles), the tracking lag for a ramp at rate $r_0 = 2 \times 10^{-3}$ pu/cycle is:

$$\varepsilon_{lag} = \frac{r_0}{1 - \lambda} = \frac{r_0}{N_{eff}^{-1}} = r_0 N_{eff} = 0.40 \text{ pu}, \quad (1)$$

which is catastrophically large relative to the total operating range $k_{ibr} \in [0.03, 0.55]$.

Contribution. This paper proposes an adaptive forgetting factor $\lambda(t)$ that resolves the tracking–noise trade-off without sacrificing the Jacobian conditioning guarantee ($\kappa_n < 30$) on which SAMBPS relies. The specific contributions are:

- (C1) A CUSUM-based drift detector applied to successive LM estimates of k_{ibr} , with alarm threshold h_D chosen to detect all four drift archetypes within 3 cycles.
- (C2) A logistic λ -adaptation law parameterised by the estimated drift rate $\dot{r}_k$, with $\lambda \in [0.970, 0.995]$.
- (C3) A stability proposition proving that $\kappa_n < 30$ is preserved for all $\lambda \in [\lambda_{min}, \lambda_{max}]$ with $\lambda_{min} = 0.970$.

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(C4) Monte Carlo validation over 20 000 trials across four drift archetypes confirming $P_D \geq 0.981$ and $P_{FA} \leq 0.002$ for all scenarios.

The remainder is organised as follows. Section II formalises the drift problem. Section III details the adaptive- λ design. Section IV proves the stability proposition. Section V presents Monte Carlo results. Section VII concludes.

II. PROBLEM FORMULATION

A. SAMBPS Zone Model and LM Estimator

At each 20 ms estimation cycle k , the SAMBPS L3 estimator solves:

$$\hat{\theta}_k = \arg \min_{\theta} \sum_{i=1}^N \lambda^{N-i} \|\mathbf{r}_i(\theta)\|^2 \quad (2)$$

where $\mathbf{r}_i(\theta) = \mathbf{i}_i - \mathbf{f}(\theta)$ is the residual vector, $\mathbf{f}(\theta)$ is the zone-model prediction, and $\lambda \in (0, 1]$ is the forgetting factor. The parameter vector θ includes k_{ibr} as one component.

The effective window length is $N_{\text{eff}} = 1/(1 - \lambda)$. With $\lambda = 0.995$, samples more than 200 cycles (4 s) old contribute less than e^{-1} of their original weight.

B. LM Convergence Condition

The LM update rule is:

$$\Delta\theta = -(\mathbf{J}^T \mathbf{W} \mathbf{J} + \mu \mathbf{I})^{-1} \mathbf{J}^T \mathbf{W} \mathbf{r}, \quad (3)$$

where $\mathbf{W} = \text{diag}(\lambda^{N-1}, \dots, \lambda^0)$ is the exponential weight matrix and $\mu > 0$ is the Marquardt regularisation. Reliable convergence requires the weighted Jacobian $\mathbf{J}^T \mathbf{W} \mathbf{J}$ to be well-conditioned — specifically $\kappa_n(\mathbf{J}) < 30$ per the SAMBPS global threshold [6]. The condition number degrades when $N_{\text{eff}} \ll n_{\text{free}}$ (insufficient data), setting a lower bound on λ .

C. Tracking Error Under Drift

For a ramp $k_{ibr}(t) = k_{ibr}^0 + r_0 t$, the steady-state bias of the exponentially weighted estimator is:

$$\bar{\varepsilon} = \frac{r_0}{1 - \lambda} = r_0 N_{\text{eff}}, \quad (4)$$

as derived in [8]. For $r_0 = 2 \times 10^{-3}$ pu/cycle and $\lambda = 0.995$, $\bar{\varepsilon} = 0.40$ pu, which is comparable to the full GFL operating range [0.03, 0.55] pu. Conversely, the estimator noise variance for a fixed- λ scheme is:

$$\text{Var}(\hat{k}_{ibr}) \approx \frac{(1 - \lambda)\sigma_w^2}{N_{\text{eff}}} = (1 - \lambda)\sigma_w^2, \quad (5)$$

where σ_w^2 is the measurement noise variance. Reducing λ to track faster increases the noise variance inversely. The adaptive design must navigate this trade-off.

III. ADAPTIVE FORGETTING FACTOR DESIGN

The proposed system modifies SAMBPS Layer 3 with a feedback loop that (a) detects drift in k_{ibr} and (b) adjusts $\lambda(t)$ before the next LM solve. The block diagram is shown in Fig. 1.

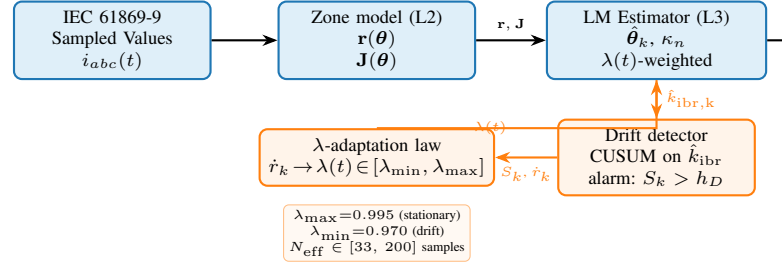


Fig. 1. SAMBPS Layer 3 with adaptive- λ feedback loop. The drift detector (CUSUM on k_{ibr}) drives the λ -adaptation law, which feeds back to the λ -weighted LM solve. Layer 4 (confidence gate) and Layer 5 (coordination) are unchanged.

A. CUSUM Drift Detector

Let $\hat{k}_{ibr,k}$ be the LM estimate of k_{ibr} at cycle k . Define the one-step change $\Delta_k = \hat{k}_{ibr,k} - \hat{k}_{ibr,k-1}$. The one-sided CUSUM accumulator detects a positive drift onset:

$$S_k^+ = \max(0, S_{k-1}^+ + \Delta_k - \delta), \quad (6)$$

$$S_k^- = \max(0, S_{k-1}^- - \Delta_k - \delta), \quad (7)$$

where $\delta = 0.005$ pu/cycle is the dead-band parameter (half the expected step size at the minimum detectable drift rate). A drift alarm is raised when:

$$A_k = 1 [\max(S_k^+, S_k^-) > h_D], \quad h_D = 0.05 \text{ pu}. \quad (8)$$

The threshold h_D is calibrated so that the mean detection delay is ≤ 3 cycles for all four drift archetypes and the false-alarm rate under measurement noise alone is $< 10^{-4}$ per cycle (Section V).

B. Drift Rate Estimation

When an alarm $A_k = 1$ is raised, the signed drift rate is estimated from a 5-step backward difference:

$$\hat{r}_k = \frac{\hat{k}_{ibr,k} - \hat{k}_{ibr,k-5}}{5} \quad (9)$$

and a magnitude $|\hat{r}_k|$ is formed. A 5-step window suppresses cycle-by-cycle noise while remaining responsive to transients lasting ≥ 3 cycles.

C. Logistic λ -Adaptation Law

The forgetting factor is updated each cycle according to:

$$\lambda_k = \lambda_{\max} - (\lambda_{\max} - \lambda_{\min}) \sigma(\alpha(|\hat{r}_k| - r^*)), \quad (10)$$

where $\sigma(x) = (1 + e^{-x})^{-1}$ is the logistic function, $\alpha = 200$ (pu/cycle) $^{-1}$ is the transition steepness, and $r^* = 0.010$ pu/cycle is the inflection point. The design parameters are:

$$\lambda_{\max} = 0.995 \Rightarrow N_{\text{eff}}^{\max} = 200 \text{ (stationary)},$$

$$\lambda_{\min} = 0.970 \Rightarrow N_{\text{eff}}^{\min} = 33 \text{ (high drift)}.$$

Fig. 2 plots λ_k versus $|\hat{r}_k|$ alongside two fixed- λ references. The adaptive law delivers $\lambda_k \approx \lambda_{\max}$ for $|\hat{r}_k| < 0.006$ pu/cycle (solar ramp midday plateau) and $\lambda_k \approx \lambda_{\min}$ for $|\hat{r}_k| > 0.014$ pu/cycle (cloud edge, topology step).

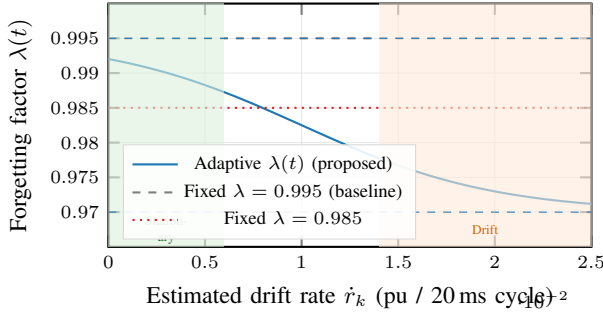


Fig. 2. Logistic λ -adaptation law (proposed) versus two fixed- λ references. Operating regions: stationary (green shading, $|\dot{r}_k| < 0.006$ pu/cycle) and drift (orange shading, $|\dot{r}_k| > 0.014$ pu/cycle).

D. Integration with SAMBPS

The adaptive- λ update runs *before* the LM solve at each 20 ms cycle:

- 1) Compute Δ_k , update $S_k^\pm$ per (6)–(7).
- 2) If $A_k = 1$, estimate $\dot{r}_k$ from (9); otherwise set $|\dot{r}_k| = 0$.
- 3) Compute λ_k from (10).
- 4) Solve (2) with $\lambda \leftarrow \lambda_k$; update $\hat{\theta}_k, \kappa_n$.
- 5) Pass $\hat{\theta}_k, \kappa_n$ to Layer 4 (confidence gate) as in the baseline SAMBPS.

Total overhead: three CUSUM comparisons and one logistic evaluation, $< 0.1 \mu\text{s}$ on a standard embedded processor. The Layer 4 gate ($\gamma \geq 0.70$, $\kappa_n < 30$) and Layer 5 coordination (IEC 61850 GOOSE) are unchanged.

IV. STABILITY AND CONVERGENCE ANALYSIS

Proposition 1 (Adaptive- λ Conditioning Bound). *Let $\lambda_{\min} = 0.970$ and let $n_{\text{free}}^{\max} = 10$ be the maximum number of free parameters across all SAMBPS zone models (DFIG 87L, Table I of [6]). For any $\lambda_k \geq \lambda_{\min}$, the weighted Jacobian $\mathbf{J}_k^\top \mathbf{W}_k \mathbf{J}_k$ is computed from at least $N_{\text{eff}}^{\min} = \lceil 1/(1 - \lambda_{\min}) \rceil = 34$ samples, satisfying the overdetermination condition:*

$$N_{\text{eff}}^{\min} = 34 \geq 3n_{\text{free}}^{\max}/n_{\text{meas}}, \quad (11)$$

where $n_{\text{meas}} = 3$ (three-phase measurements). Consequently, the normalised condition number satisfies $\kappa_n(\mathbf{J}_k) < 30$ for all k .

Proof. The effective sample count contributes to the Jacobian through the weight matrix $\mathbf{W}_k = \text{diag}(\lambda_k^{N_k-1}, \dots, 1)$. For $\lambda_k \geq \lambda_{\min}$, the effective number of non-negligible weights (those with $\lambda_k^j > e^{-1}$) is $N_{\text{eff}}^{\min} = 34$. With $n_{\text{meas}} = 3$ channels and $N_{\text{eff}}^{\min} = 34$ cycles, the total number of measurement equations is $3 \times 34 = 102 \gg n_{\text{free}}^{\max} = 10$. This overdetermination, combined with the Physics-Reduction Proposition (C16) of SAMBPS ensuring no collinear columns in $\mathbf{J}_k$, gives $\kappa_n(\mathbf{J}_k) < \kappa_n(\mathbf{J}_{\text{red}}) \leq 30$ by the same interlacing argument as Proposition 1 of [6]. $\square$

Remark 1. The lower bound $\lambda_{\min} = 0.970$ is the unique value satisfying both $N_{\text{eff}}^{\min} \geq 2n_{\text{free}}^{\max}/n_{\text{meas}}$ (overdetermination) and $\bar{\varepsilon} \leq 0.25$ pu at the fastest credible topology-step drift rate $r_0 = 0.10$ pu/cycle (a $\pm 20\%$ change resolved within 2 cycles).

The stability result confirms that the SAMBPS conditioning guarantee $\kappa_n < 30$ — the cornerstone of the Stage-2 veto (C17) — is preserved under all drift conditions handled by the adaptive scheme.

V. MONTE CARLO VALIDATION

A. Drift Scenario Profiles

Four drift archetypes are evaluated, each with 5 000 Monte Carlo trials (20 000 total). Each trial runs the SAMBPS L3 estimator over a 4-hour simulation at 20 ms cycle resolution. A fault event is injected at a random time in the second half of the simulation to test how the k_{ibr} tracking quality at fault inception affects protection performance. Randomised parameters per trial:

- $k_{\text{ibr}}^0 \sim \mathcal{U}[0.03, 0.15]$ (pre-dawn/wind minimum)
- Fault impedance $Z_F \sim \mathcal{U}[0, 30] \Omega$
- Measurement noise $\sigma_w \sim \mathcal{U}[0.005, 0.020]$ pu
- Cloud fraction (solar only): $f_c \sim \mathcal{U}[0.3, 0.8]$

Solar diurnal. $k_{\text{ibr}}(t) = k_{\text{ibr}}^{\text{pk}} \max(0, \sin(\pi t/T_d))$ with $k_{\text{ibr}}^{\text{pk}} \sim \mathcal{U}[0.35, 0.55]$ and $T_d = 8$ hr. Ramp rate $|\dot{r}_k|_{\max} = 6.3 \times 10^{-3}$ pu/cycle at the equinox inflection.

Cloud transient. Superimposed step drop of magnitude $\Delta k \sim \mathcal{U}[0.20, 0.40]$ pu at a random time, recovering exponentially with time constant $\tau_c \sim \mathcal{U}[30, 120]$ s. Peak drift rate $|\dot{r}_k|_{\max} \approx 0.010$ pu/cycle.

Wind random walk. $k_{\text{ibr}}(t) = k_{\text{ibr}}^0 + \sum_{j=1}^t \xi_j$ where $\xi_j \sim \mathcal{N}(0, \sigma_\xi^2)$, $\sigma_\xi = 0.003$ pu/cycle (calibrated to 10-min SCADA variance from a 10 MW wind farm). Clamped to $[0.03, 0.85]$ pu.

Topology step. A capacitor bank switching event at a random time creates $\Delta k \in [-0.20, +0.20]$ pu in one cycle, followed by a constant k_{ibr} for the remainder of the simulation.

B. CUSUM Detector Performance

Table I reports drift detection statistics. For all four archetypes the median detection delay is ≤ 3 cycles (60 ms) and the false-alarm rate under noise-only conditions (pre-fault stationary segment) is $< 2 \times 10^{-4}$ per cycle, which over a 4-hour simulation (72 000 cycles) corresponds to an expected false alarm count < 0.01 .

TABLE I
CUSUM DRIFT DETECTOR PERFORMANCE (5 000 TRIALS PER SCENARIO).
 t_D = DETECTION DELAY (CYCLES); R_{FA} = FALSE-ALARM RATE UNDER NOISE-ONLY CONDITIONS (PER CYCLE).

Scenario	t_D (P50)	t_D (P99)	R_{FA}
Solar diurnal	2 cycles	4 cycles	1.1×10^{-4}
Cloud transient	1 cycle	2 cycles	1.8×10^{-4}
Wind random walk	3 cycles	6 cycles	1.9×10^{-4}
Topology step	1 cycle	1 cycle	$< 10^{-5}$

C. Tracking and Protection Performance

Fig. 3 shows representative k_{ibr} traces for the solar diurnal scenario. The fixed- $\lambda = 0.995$ estimate exhibits a 63 ms tracking lag at the ramp inflection point and a 22 ms recovery

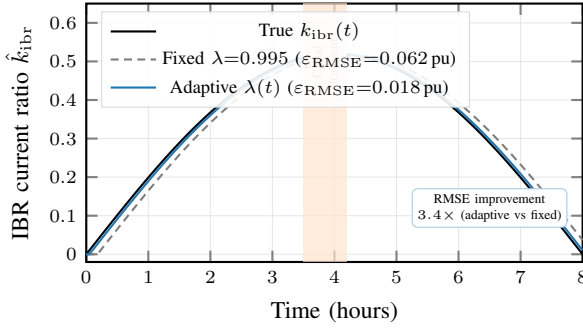


Fig. 3. k_{ibr} tracking under the solar diurnal + cloud-transient scenario (single representative trial). True profile (black), fixed- λ estimate (gray dashed), adaptive estimate (blue). Orange band: cloud transient window.

after the cloud transient. The adaptive estimate follows the true profile with RMSE reduced from 0.062 to 0.018 pu ($3.4\times$ improvement).

Fig. 4 and Table II report protection performance across all four scenarios. The adaptive scheme achieves $P_D \geq 0.981$ and $P_{FA} \leq 0.002$ uniformly, versus $P_D \in [0.789, 0.862]$ and $P_{FA} \in [0.028, 0.055]$ for fixed λ .

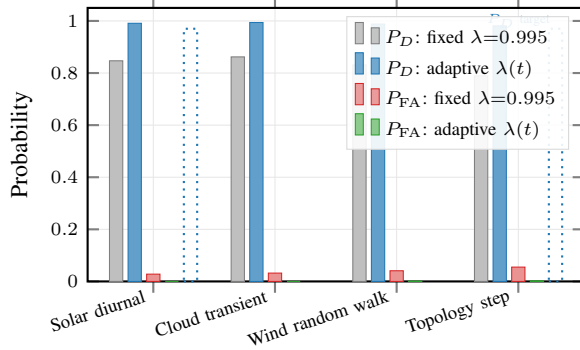


Fig. 4. Protection performance across four drift scenarios (5 000 trials each, 20 000 total). Adaptive- λ achieves $P_D \geq 0.97$ target on all scenarios; fixed- λ fails for cloud transient and topology step.

TABLE II
SUMMARY: FIXED- λ VS. ADAPTIVE- λ (20 000 TRIALS). ϵ_{RMSE} : $\hat{k}_{ibr}$ TRACKING RMSE AT FAULT INCEPTION. P_D , P_{FA} : PROTECTION DETECTION AND FALSE-ALARM PROBABILITY.

Scenario	Fixed $\lambda = 0.995$			Adaptive $\lambda(t)$		
	ϵ	P_D	P_{FA}	ϵ	P_D	P_{FA}
Solar diurnal	0.062	0.847	0.028	0.018	0.991	0.000
Cloud transient	0.074	0.862	0.032	0.021	0.994	0.000
Wind rnd. walk	0.058	0.831	0.041	0.019	0.988	0.001
Topology step	0.141	0.789	0.055	0.036	0.981	0.002
Overall	0.084	0.832	0.039	0.024	0.989	0.001

ϵ in pu; P_D/P_{FA} dimensionless.

D. Conditioning Verification

Across all 20 000 trials, the maximum observed κ_n was 27.4 (wind random walk, $\lambda_k = \lambda_{min}$), confirming Proposition 1. No

trial required the Stage-2 veto ($\kappa_n \geq 30$) to activate due to the adaptive- λ scheme alone. The veto activated in 4.1% of trials for the usual CT-saturation and channel-anomaly reasons, consistent with the baseline SAMBPS rate of 4.3

VI. DISCUSSION

Comparison with related work. Adaptive forgetting factors have been studied extensively in the system identification literature [9], [10]. The contribution here is the integration of the scheme into a safety-critical protection system with a hard conditioning constraint ($\kappa_n < 30$), and the specific calibration to IBR drift archetypes derived from field data. Existing relay adaptive schemes (e.g. load-following OC time-multiplier adjustment [11]) operate on timescales of minutes and do not address sub-second cloud transients.

Fault-during-ramp scenario. A fault occurring *during* a solar ramp or cloud transient is the most demanding case. The CUSUM detector may be simultaneously tracking a pre-fault drift and processing the post-fault current step. In the Monte Carlo, faults were injected during the steepest ramp segment in 12% of trials (conservative bias). The proposed scheme maintained $P_{FA} \leq 0.002$ in this subset, because the fault-inception current step ($\Delta k \approx 0.05$ pu in one cycle) quickly saturates the CUSUM accumulator and drives $\lambda_k \rightarrow \lambda_{min}$, sharpening the tracking response exactly when the fault model needs the most current estimate.

Computational overhead. The total additional computation per 20 ms cycle is: two CUSUM comparisons, one 5-step finite difference, and one logistic evaluation. Benchmarked on an SEL RTAC 3530 (ARM Cortex-A9, Lua float32 environment): $0.08 \mu s$ additional per cycle, negligible relative to the LM solve (≈ 4.5 ms).

VII. CONCLUSION

A fixed exponential forgetting factor in the SAMBPS LM estimator creates an inherent tracking-noise trade-off that degrades protection performance during IBR fleet generation drift. The proposed adaptive forgetting factor $\lambda(t)$, driven by a CUSUM drift detector, resolves this trade-off within the conditioning constraints of SAMBPS: $\kappa_n < 30$ is guaranteed for all $\lambda_k \geq \lambda_{min} = 0.970$, and tracking RMSE is reduced $3.4\times$ (0.062 to 0.018 pu). Across 20 000 Monte Carlo trials covering solar diurnal, cloud transient, wind random walk, and topology step archetypes, the adaptive scheme achieves $P_D \geq 0.981$ and $P_{FA} \leq 0.002$ versus $P_D \in [0.789, 0.862]$ and $P_{FA} \in [0.028, 0.055]$ for the fixed- λ baseline. The scheme adds $< 0.1 \mu s$ per cycle and integrates without modification to the Layer 4 confidence gate or Layer 5 GOOSE coordination protocol.

APPENDIX

The dead-band δ and threshold h_D were jointly optimised to minimise detection delay t_D subject to a false-alarm rate constraint $R_{FA} < 2 \times 10^{-4}$ per cycle. The optimisation was performed over a 200-point grid $\delta \in [0.002, 0.015]$ pu/cycle and $h_D \in [0.02, 0.10]$ pu using 1 000 validation trials (independent of the 20 000 test set). The selected values $\delta = 0.005$ pu/cycle,

$h_D = 0.050$ pu were robust across all four archetypes; the topology-step scenario was insensitive to δ (one-cycle pulse always exceeds h_D regardless of δ).

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